Name: _____

Student ID: _____

Campus: _____

CS310 – Computer Networks

Semester I, 2024
Mode: Face to Face & Remote Students

Short Test 1

Date: Friday 4th April 2024; Time: 9.00 –9.55 PM (FJ Time)
Weight: 10%

You have 5 minutes to read this paper,

Followed by 50 minutes to complete this paper.

All Questions are Compulsory

This Test 1 will focus on the following learning outcomes for this course:

CLO1. Differentiate the roles of various protocols in the TCP/IP protocol stack

CLO3. Assess networked systems in terms of performance metrics such as throughput, latencies, and resource utilization

Core Body of Knowledge		CS310	Test 1
	Complex Computing		
ICT Professional Knowledge	Ethics		
	Professional expectations		
	Teamwork concepts/issues		
	Communication		
ICT Problem Solving:	Abstraction		
	Design	M	▯
Technology Resources	Hardware and Software Fundamentals		
	Data and Information Management		
	Networking	M	▯
Technology Building	Human Factors		
	Programming		
	Systems Development / Acquisition		
ICT Management	IT Governance and organizational issues		
	IT Project management		
	Service management		
	Security management		

This Test 1 consists of three sections:

1. Section A: 10 Marks
2. Section B: 10 Marks 3. Section C: 10 Marks

*Zero for any Plagiarism case.

Question 1:

i Wireshark serves as a prominent network protocol analyzer, enabling users to capture and dynamically explore the data traffic within a computer network. Applying the insights gained from your lab sessions, identify and label the five crucial components of the Wireshark interface.

Hint: *Use arrows within the provided Figure 1 to indicate the precise locations and assign appropriate labels to each of five components.* [2.5 marks]

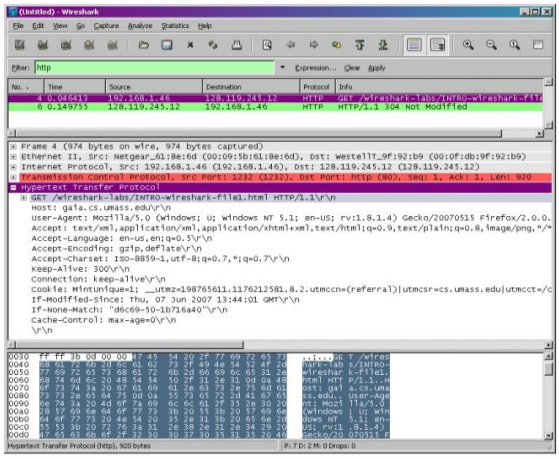
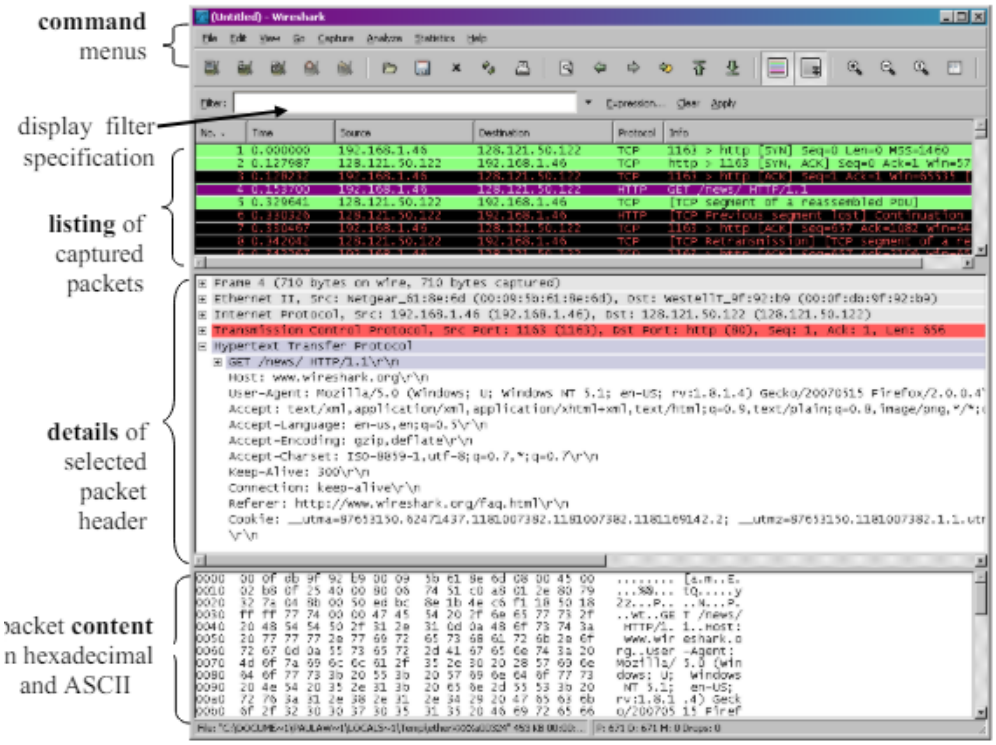
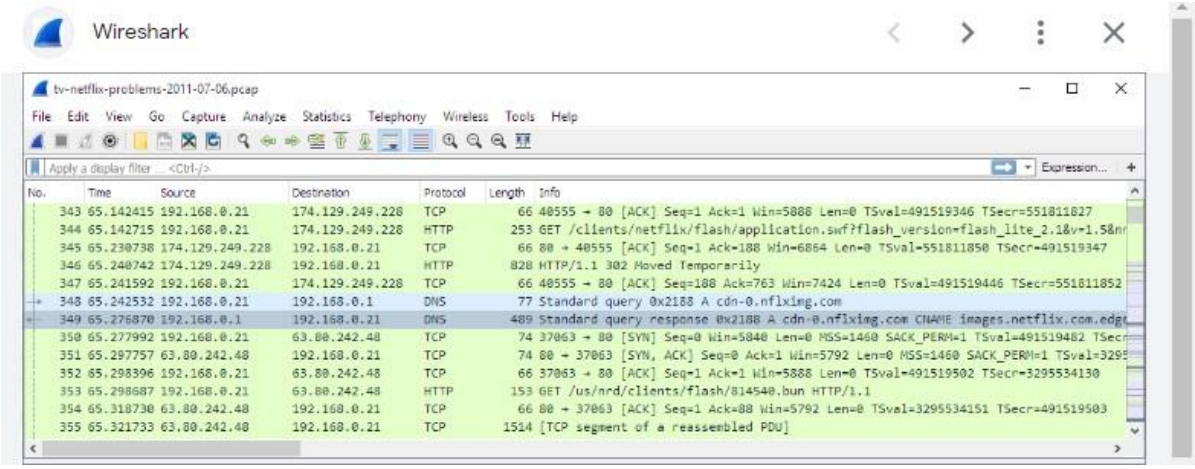


Figure 1. Wireshark display.



ii Figure 2 below presents Wireshark capture when a long HTML file is downloaded from a site. [1.5 marks]



a. State the number of packets that contain GET message: 2

b. The number of packets that contains the status code and phrase associated with response to the HTTP GET request is 0

c. What does the presence of an HTTP/1.1 302 Moved Temporarily message in Wireshark? signify? *Hint: Use the figure 2 above [line 346].*

1. **HTTP/1.1**: This is the version of the HTTP protocol being used. HTTP/1.1 is a widely used version of the protocol.

2. **302 Moved Temporarily**: This is an HTTP status code. It means that the requested resource has been temporarily moved to a different location. The client should continue to use the original URL for future requests, but they should also use the new URL provided in the response for this specific request.

When you see this message in Wireshark, it indicates that a web server has responded to a client's request by saying, "Hey, the resource you're looking for has moved, but only temporarily. Here's where you can find it for now." This could happen for various reasons, such as server maintenance, load balancing, or content reorganization.

Question 2:

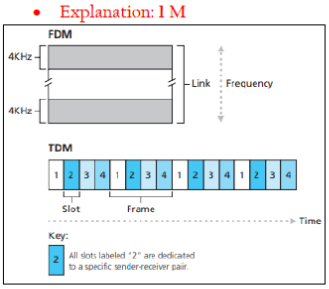
The Circuit switched links involve either Frequency Division Multiplexing [FDM] or Time Division Multiplexing [TDM].

Sketch FDM and TDM for a specific network link supporting up to **four circuits**. Explain how the bandwidth is divided in both cases.

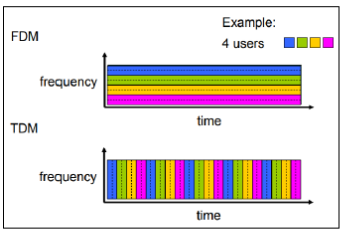
- Ensure to clearly show:

The Link Frequency [FDM] and Slot, Frame & Time [TDM] [2 Marks]

- Accepted Both [1 M for FDM, 1 M for TDM]
- The Link Frequency [FDM] and Slot, Frame & Time [TDM] – 1M
- Explanation: 1 M



OR



With FDM, each circuit continuously gets a fraction of the bandwidth. With TDM, each circuit gets all of the bandwidth periodically during brief intervals of time (that is, during slots) or any similar explanation is accepted.

Question 3:

Recall that a packet starts in a host (the source), passes through a series of routers, and ends its journey in another host (the destination). As a packet travels from one node (host or router) to the subsequent node (host or router) along this path, the packet suffers from several types of delays at each node along the path.

Clearly label all delays that may exist and identify their position in the diagram given

Hint: Draw and label in figure 3 below. Ensure to clearly highlight the areas. [1.5 Marks]

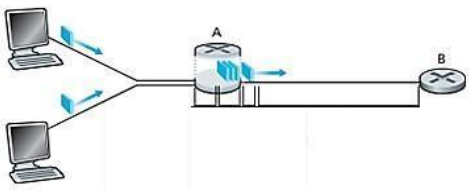


Figure 1: A model that portrays various types of delays.

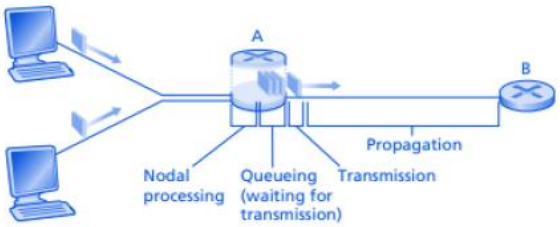
Solution

Count for only:

Queuing Delay [0.5 M]

Transmission Delay and [0.5 M]

Propagation Delay [0.5 M]



Question 4:

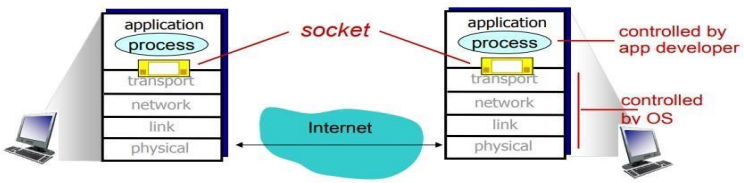


Figure 2: A the typical flow of events for a connection-oriented socket.

The above diagram shows the typical flow of events for a connection-oriented socket.

a. Describe in detail the analogy of sockets, you need to mention the typical flow of events between the application layer and the transport layer. [1.5 marks]

The process sends/receives messages to/from its socket. The socket analogous to door; the sending process shoves message out door. Remember the sending process relies on transport infrastructure on other side of door to deliver message to socket at receiving process

For example:

The server socket listens for incoming connections. A server creates a socket, binds the socket to an IP address and port number (for TCP and UDP), and then listens for incoming connections. When a client connects to the server, a new socket is created for communication with the client (TCP only).

OR

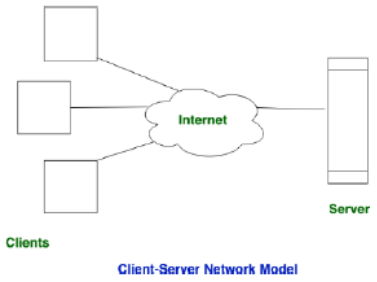
Any relevant explanation is excepted.

b. Applications with Peer-to-Peer [P2P] architectures have client processes & server processes. Differentiate and explain the P2P and Client to Server processes. [1 mark]

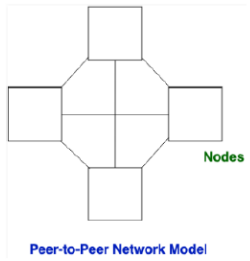
Client-Server Network:

This model is broadly used network model. In Client-Server Network, Clients and server are differentiated, Specific server and clients are present. In Client-Server Network, Centralized server is used to store the data because its management is centralized.

In Client-Server Network, Server respond the services which is request by Client.



Peer-to-Peer Network: This model does not differentiate the clients and the servers, In this each and every node is itself client and server. In Peer-to-Peer Network, Each and every node can do both request and respond for the services.



OR

Any relevant explanation is excepted.

SECTION B – The Web and HTTP [10 Marks]

Question 5:

a. Consider the two 16-bit words (shown in binary) below. Recall that to compute the Internet checksum of a set of 16-bit words, we compute the one's complement sum [1] of the two words. That is, we add the two numbers together, making sure that any carry into the 17th bit of this initial sum is added back into the 1's place of the resulting sum); we then take the one's complement of the result. [3 marks]

00001111 11001011 this binary number is 4043 decimals (base 10)

11111100 11001111 this binary number is 64719 decimals (base 10)

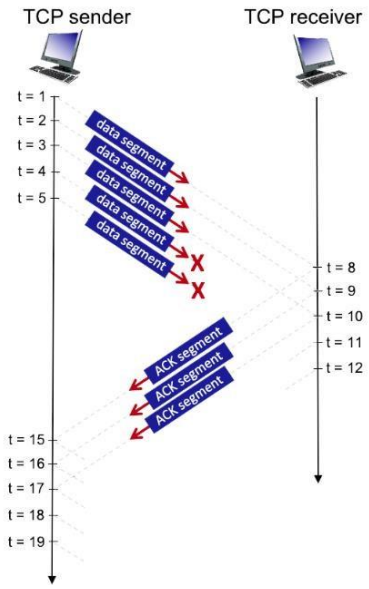
Compute the Internet checksum value for these two 16-bit words: Show detailed working.

For the binary number 00001111 11001011 (4043 in decimal):a. Add the two 16-bit words:
0000111111001011 (binary)
+ 1111110011001111 (binary)
~~~~~~~~~~~~~~~~~~
0000110010011010 (binary)

One's complement of 0000110010011010

=**1111001101100101**
~~~~~~~~~~~~~~~~~~

b. Consider the figure below in which a TCP sender and receiver communicate over a connection in which the segments can be lost. [7 marks]



a. What is the sequence number of the segment sent at t=1, 2, 3, 4 and 5?

- Initial sequence number sent: 104
- Segment size sent: 174

  At time t=0, the sequence number of the segment sent is 104.

  For subsequent time intervals:
- **At t=1, the sequence number would be 104 + 174 = 278.**
- **At t=2, the sequence number would be 278 + 174 = 452.**
- **At t=3, the sequence number would be 452 + 174 = 626.**
- **At t=4, the sequence number would be 626 + 174 = 800.**
- **At t=5, the sequence number would be 800 + 174 = 974.**

  So, the sequence numbers of the segments sent at t=1, 2, 3, 4, and 5 are 278, 452, 626, 800, and 974, respectively.

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b. What is the value of the ACK sent at t=8? (If segment lost, write 'x')

At t=8, the sequence number of the segment sent is 104 (initial sequence number) + 7 * 174 (assuming segments were sent every time unit). This gives us a sequence number of 1358.

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c. What is the sequence number of the segment sent at t = 19? (If ACK never arrives, write 'x')

Since the segment size is 174 and 19 time units have passed, we can calculate the sequence number as follows:

Sequence number at t=19 = Initial sequence number + (Number of segments sent * Segment size)

Sequence number at t=19 = 104 + (19 * 174)

Sequence number at t=19 = 104 + 3306

Sequence number at t=19 = 3410

Therefore, the expected sequence number of the segment sent at t=19 is 3410.

SECTION C – The Web and HTTP [20 Marks]

Question 6:

An FTP client requires the IP address of an FTP server. Suppose in the path to resolving the IP address, there are four DNS servers (Local DNS server, root DNS server, TLD DNS server, authoritative DNS server).

a. Assuming the cache of the client's local DNS server is empty, how many round trips (RTTs) would it take for the FTP client to get the IP address of the FTP server using iterative DNS query and ii) recursive DNS query. [2 marks]

i RTTs for iterative DNS query_____.

ii RTTs for recursive DNS query_____.

a. **Iterative DNS Query:**

In an iterative DNS query, the client directly queries each DNS server in sequence until it reaches the authoritative DNS server. Let's break down the steps:

1. The FTP client sends a query to the local DNS server.
2. The local DNS server, which doesn't have the IP address cached, forwards the query to a root DNS server.
3. The root DNS server replies with the IP address of the TLD DNS server responsible for the top-level domain (e.g., .com, .net).
4. The local DNS server then queries the TLD DNS server.
5. The TLD DNS server responds with the IP address of the authoritative DNS server for the domain.
6. Finally, the local DNS server queries the authoritative DNS server and receives the IP address of the FTP server.

So, in total, there are 4 RTTs for an iterative DNS query.

b. **Recursive DNS Query:**

In a recursive DNS query, the client delegates the entire resolution process to the DNS server, which resolves the IP address recursively on behalf of the client.

1. The FTP client sends a query to the local DNS server.
2. The local DNS server recursively queries each DNS server in sequence (root, TLD, authoritative) until it directly obtains the IP address of the FTP server from the authoritative DNS server.
3. The local DNS server then replies to the FTP client with the IP address.

So, in total, there are 2 RTTs for a recursive DNS query.

b. Now Assume that the cache of the client's local DNS server contains the requested IP address, how many RTT will it take for the client to get the IP address. [1 mark]

If the requested IP address is already cached in the client's local DNS server, the resolution process will be expedited as there is no need to query other DNS servers. Therefore, regardless of whether it's an iterative or recursive query, it will only take one round trip (RTT) for the client to retrieve the IP address from its local DNS cache.

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c. DNS runs over which transport layer protocol. [1 mark]

DNS (Domain Name System) typically runs over the User Datagram Protocol (UDP) on port 53. However, it can also use Transmission Control Protocol (TCP) when the response data size exceeds the maximum limit for UDP or for other specific purposes such as zone transfers. So, DNS can utilize both UDP and TCP as the transport layer protocols.

d. Transport layer protocols facilitate process-to-process communication running on hosts over the network. List two fields of information that uniquely identifies such a process. [1 mark]

Two fields of information that uniquely identify a process in process-to-process communication over the network are:

- Port Number: Port numbers are used to distinguish between different processes running on the same host. They range from 0 to 65535 and help route incoming data packets to the correct process.
- IP Address: The IP address of the host identifies the specific device on the network. When combined with the port number, it forms a unique address for the process.

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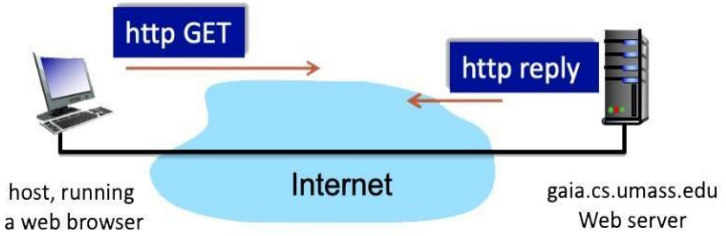
e. Differentiate between TCP flow control and congestion control. [2 marks]

TCP flow control regulates the rate of data transmission to match the receiver's ability to process data, while TCP congestion control regulates the rate of data transmission to prevent network congestion and ensure efficient utilization of network resources.

Question 7:

Consider the figure below, where the server is sending a HTTP RESPONSE message back the client. Suppose the server-to-client HTTP RESPONSE message is the following:

HTTP/1.0 200 OK
Date: Thu, 28 Apr 2022 00:00:48 +000 0
Server: Apache/2.2.3 (CentOS)
Last-Modified: Wed, 27 Apr 2022 23:58:28 +0000
ETag:17dc6-a5c-bf716880.
Content-Length: 740
Connection: Close
Content-type: text/html



i) Clearly identify the response message and justify on the document status which was sent from client to server. [3 marks]

HTTP/1.0; Evidences show that the response message to the client is a success!

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ii Describe with appropriate examples when should you use a User Datagram Protocol (UDP) instead of Transmission Control Protocol (TCP)? [3 marks]

UDP (User Datagram Protocol) UDP (User Datagram Protocol) is an alternative communications protocol to Transmission Control Protocol (TCP) used primarily for establishing low-latency and loss-tolerating connections between applications on the internet.

UDP is commonly used for streaming audio and video.

Streaming media, real-time multiplayer games and voice over IP (VoIP) are examples of applications that often use UDP. In these particular applications, loss of packets is not usually a fatal problem.

OR

Any relevant explanation is excepted.

Question 7: The Web and HTTP

i) Non-persistent HTTP! suppose user enters URL: www.usp.edu/someDepartment/home.index

a. Contains the references to 25 jpg images.

Describe in complete steps what happens when the object request [25] is sent over a connection.

Hint: Ensure to remark on the connection type and port # as well. [4 marks]

STEP1. 1a. HTTP client initiates TCP connection to HTTP server (process) at www.someSchool.edu on port 80
1b. HTTP server at host www.someSchool.edu waiting for TCP connection at port 80. "accepts" connection, notifying client

STEP2. HTTP client sends HTTP request message (containing URL) into TCP connection socket. Message indicates that client wants object someDepartment/home.inde x

STEP3. HTTP server receives request message, forms response message containing requested object, and sends message into its socket

STEP4. HTTP server closes TCP connection.

STEP5. HTTP client receives response message containing html file, displays html. Parsing html file, finds 10 referenced jpeg objects

STEP6. Steps 1-5 repeated for each of 10 jpeg objects

Question 8: The HTTP Request to A Web Server

```
GET /hr/staff.html HTTP/1.1
Date: Wed, 20 Apr 2011 14:12:31 PDT
If-Modified-Since: Mon, 13 Apr 2011 19:43:31 PDT
Host: www.usp.ac.fj
User-agent: Mozilla/4.0
Connection: Keep-Alive
Accept: text/html, text/plain, image/gif, image/jpeg Accept-
language: en
```

a. What is the difference between persistent and non-persistent connection. Is the above connection request persistent or non-persistent? [2 marks]

1. **Persistent Connection:**

- In a persistent connection, the connection between the client and server remains open after the initial request and response are completed.
- Multiple requests and responses can be sent over the same connection without the need to establish a new connection for each request.
- Persistent connections reduce the overhead of establishing and tearing down connections for each request, leading to better performance and reduced latency, especially for websites with multiple resources to load.
- Persistent connections are often used in modern web applications and HTTP/1.1 or later versions by default unless explicitly disabled.

2. **Non-persistent Connection:**

- In a non-persistent connection, the connection is closed after each request-response cycle.
- For each new request, the client establishes a new connection to the server, sends the request, receives the response, and then closes the connection.
- Non-persistent connections result in higher overhead due to the need to establish and tear down connections repeatedly, potentially leading to slower performance and increased latency, especially for websites with multiple resources requiring separate connections.

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b. Suppose the page (staff.html) contains reference to 5 images. How many round trips (RTT) are needed to get the page and the 5 images from the web server using persistent and non-persistent connection. (Ignore closing down of connection). [1 mark]

- Each resource (HTML page and images) requires one RTT to request and receive.
- Persistent connections allow multiple resources to be retrieved over the same connection without additional RTTs, while non-persistent connections require a separate connection for each resource.

Persistent Connection:

1. One RTT to request and receive the HTML page (**staff.html**).
2. Since persistent connections are used, no additional RTTs are needed for the 5 images; they can be requested and received over the same open connection.

Total RTTs = 1 (for HTML page)

Non-Persistent Connection:

1. One RTT to request and receive the HTML page (**staff.html**).
2. Five additional RTTs for each image, as non-persistent connections require a separate connection for each resource.

Total RTTs = 1 (for HTML page) + 5 (for images) = 6

Therefore, using a persistent connection requires 1 RTT, while using a non-persistent connection requires 6 RTTs to retrieve the HTML page and the 5 images.

The End